

Experiment No – 2.A

Objective – To implement Linear Search.

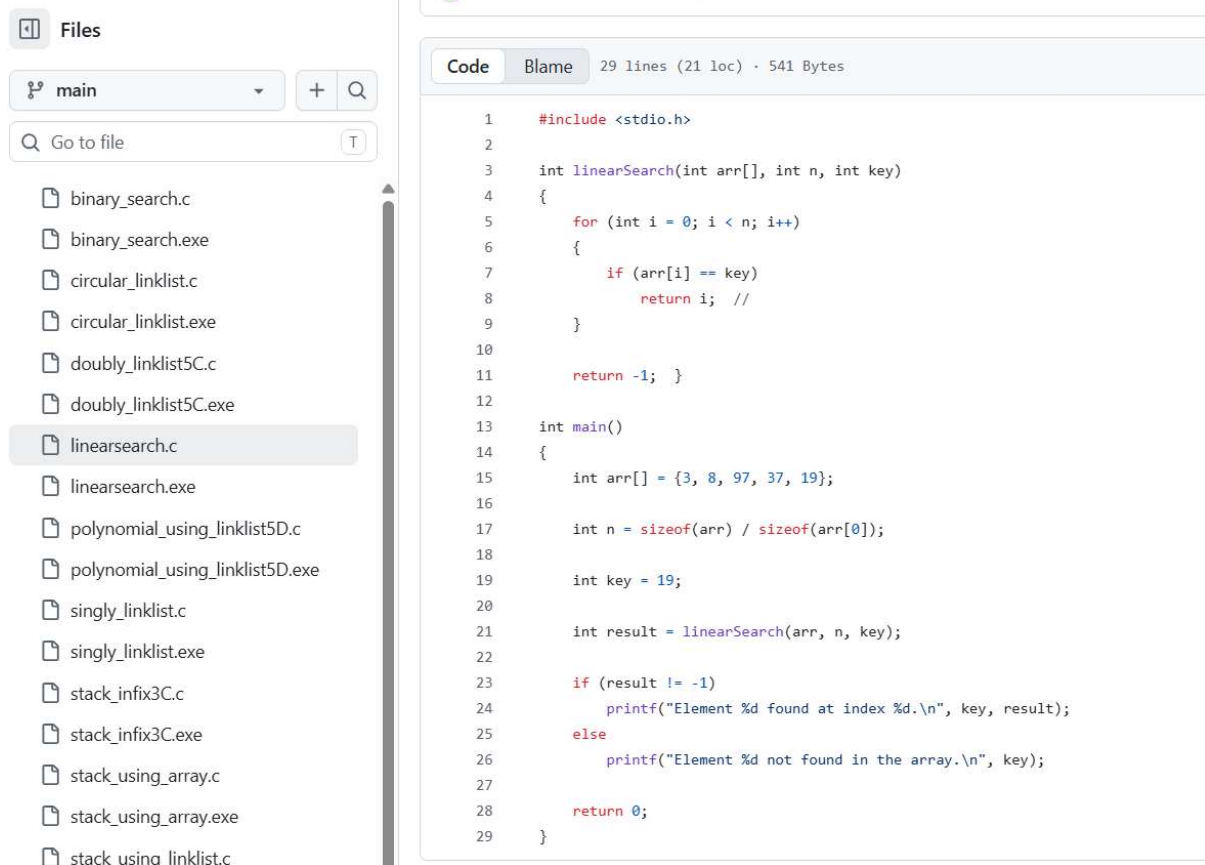
Input :

The screenshot displays a GitHub repository for 'alokkumar75 / DS-Lab'. The 'Code' tab is selected, showing the file 'linearsearch.c'. The file content is as follows:

```
1  #include <stdio.h>
2
3  int linearSearch(int arr[], int n, int key)
4  {
5      for (int i = 0; i < n; i++)
6      {
7          if (arr[i] == key)
8              return i; //
9      }
10
11     return -1; }
12
13 int main()
14 {
15     int arr[] = {3, 8, 97, 37, 19};
16
17     int n = sizeof(arr) / sizeof(arr[0]);
18
19     int key = 19;
20
21     int result = linearSearch(arr, n, key);
22
```

The left sidebar shows a list of files in the repository, with 'linearsearch.c' highlighted. The top navigation bar includes links for Code, Issues, Pull requests, Agents, Actions, Projects, and Security and quality.

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Files

main

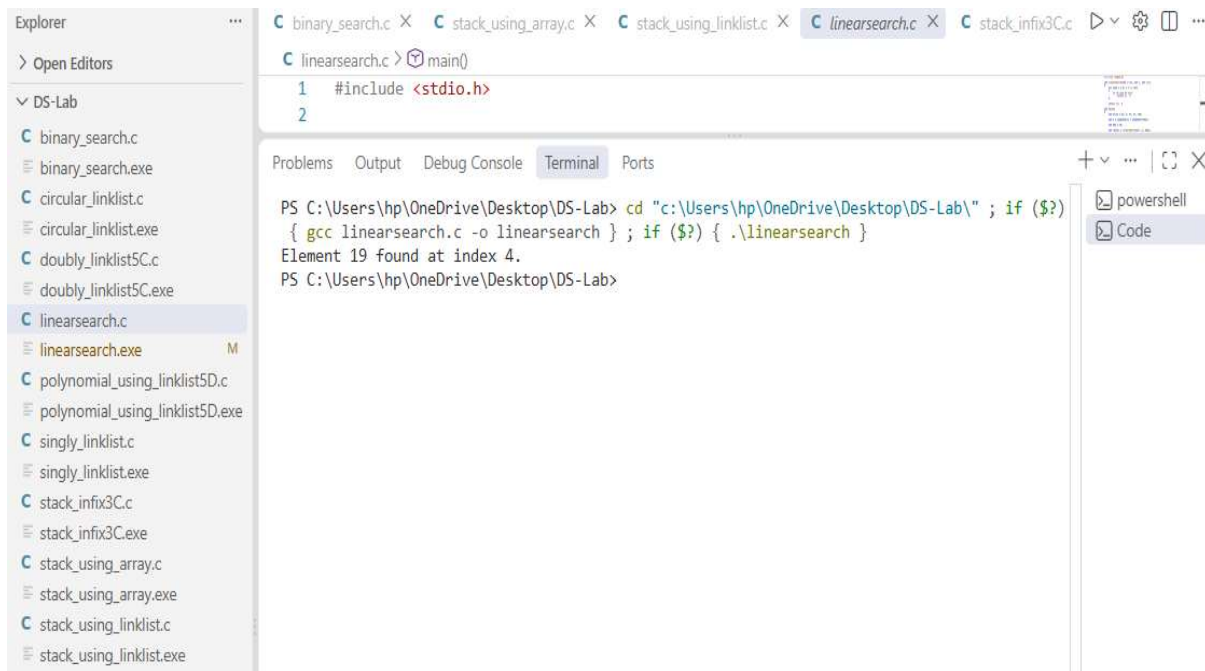
Go to file

- binary_search.c
- binary_search.exe
- circular_linklist.c
- circular_linklist.exe
- doubly_linklist5C.c
- doubly_linklist5C.exe
- linearsearch.c**
- linearsearch.exe
- polynomial_using_linklist5D.c
- polynomial_using_linklist5D.exe
- singly_linklist.c
- singly_linklist.exe
- stack_infix3C.c
- stack_infix3C.exe
- stack_using_array.c
- stack_using_array.exe
- stack_using_linklist.c

Code Blame 29 lines (21 loc) · 541 Bytes

```
1  #include <stdio.h>
2
3  int linearSearch(int arr[], int n, int key)
4  {
5      for (int i = 0; i < n; i++)
6      {
7          if (arr[i] == key)
8              return i; //
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11     return -1; }
12
13 int main()
14 {
15     int arr[] = {3, 8, 97, 37, 19};
16
17     int n = sizeof(arr) / sizeof(arr[0]);
18
19     int key = 19;
20
21     int result = linearSearch(arr, n, key);
22
23     if (result != -1)
24         printf("Element %d found at index %d.\n", key, result);
25     else
26         printf("Element %d not found in the array.\n", key);
27
28     return 0;
29 }
```

Output :



Explorer

Open Editors

DS-Lab

- binary_search.c
- binary_search.exe
- circular_linklist.c
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- stack_infix3C.exe
- stack_using_array.c
- stack_using_array.exe
- stack_using_linklist.c
- stack_using_linklist.exe

linearsearch.c > main()

```
1  #include <stdio.h>
2
```

Problems Output Debug Console **Terminal** Ports

PS C:\Users\hp\OneDrive\Desktop\DS-Lab> cd "c:\Users\hp\OneDrive\Desktop\DS-Lab" ; if (\$?) { gcc linearsearch.c -o linearsearch } ; if (\$?) { .\linearsearch }
Element 19 found at index 4.
PS C:\Users\hp\OneDrive\Desktop\DS-Lab>

powershell

Code

Experiment No – 2.A

The screenshot displays the Visual Studio Code interface. On the left, the Explorer sidebar shows a project named 'DS-Lab' containing various C source files and their corresponding executables. The file 'linearsearch.c' is selected. The main editor window shows the code for 'linearsearch.c', which includes `<stdio.h>` and a `main()` function. Below the editor, the Terminal panel is active, showing the command prompt output. The user has navigated to the directory 'c:\Users\hp\OneDrive\Desktop\DS-Lab' and compiled the program using `gcc linearsearch.c -o linearsearch`. The program has been executed, and the output shows 'Element 19 found at index 4'.

Explorer

Open Editors

DS-Lab

- binary_search.c
- binary_search.exe
- circular_linklist.c
- circular_linklist.exe
- doubly_linklist5C.c
- doubly_linklist5C.exe
- linearsearch.c
- linearsearch.exe
- polynomial_using_linklist5D.c
- polynomial_using_linklist5D.exe
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- singly_linklist.exe
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- stack_infix3C.exe
- stack_using_array.c
- stack_using_array.exe
- stack_using_linklist.c
- stack_using_linklist.exe

linearsearch.c > main()

```
1 #include <stdio.h>
2
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\hp\OneDrive\Desktop\DS-Lab> cd "c:\Users\hp\OneDrive\Desktop\DS-Lab\" ; if ($?) {
  gcc linearsearch.c -o linearsearch } ; if ($?) { .\linearsearch }
Element 19 found at index 4.
PS C:\Users\hp\OneDrive\Desktop\DS-Lab>
```

powershell

Code